

Continuous Random Variables

Reading Objectives:

In this reading, you will be introduced to the probability density function of a continuous random variable and its mean, variance, and standard deviation with examples.

Main Reading Section:

Definition 1: We say that X is a continuous random variable if there exists a nonnegative function $f(x)$, defined for all real $x \in (-\infty, \infty)$, having the property that for any set B of real numbers

$$P(X \in B) = \int_B f(x) dx$$

The function $f(x)$ is called the **probability density function** (PDF) of the random variable X .

Since X must assume some value, $f(x)$ must satisfy $\int_{-\infty}^{\infty} f(x) dx = 1$.

$$P\{a \leq X \leq b\} = \int_a^b f(x) dx$$

$$P\{X = a\} = 0$$

The relationship between the cumulative distribution $F(\cdot)$ and the probability density $f(\cdot)$ is expressed by $F(a) = P\{X \in (-\infty, a]\} = \int_{-\infty}^a f(x) dx$. This also implies $\frac{d}{da} F(a) = f(a)$.

Variance and Standard deviation

Definition 2: If X is a random variable with mean μ , then the variance of X , denoted by $\text{Var}(X)$, is defined by

$$\text{Var}(X) = E[(X - \mu)^2] \quad \text{or} \quad \text{Var}(X) = E[X^2] - E[X]^2$$

Definition 3: The quantity $\sqrt{\text{Var}(X)}$ is called the standard deviation of X .

Properties of variance:

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

Example 1: Let X be a random variable with PDF given by

$$f(x) = \begin{cases} \frac{3}{2}x^2, & |x| \leq 1 \\ 0, & \text{Otherwise} \end{cases}$$

. Find $E[X]$ and $\text{Var}(X)$.

$$. P(X \geq \frac{1}{2})$$

Solution:

$$. E[X] = \int_{-1}^1 w f(w) dw = \frac{3}{2} \int_{-1}^1 w^3 dw = 0$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = \int_{-1}^1 w^2 f(w) dw = \frac{3}{2} \int_{-1}^1 w^4 dw = \frac{3}{5}$$

$$. P(X \geq \frac{1}{2}) = \frac{3}{2} \int_{1/2}^1 w^2 dw = \frac{7}{16}$$